Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-8)

COMPUTER AIDED DRUG DESIGN

Subject Code : BP807ET

M.Code : 79770

Date of Examination : 24-04-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A Is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Answer briefly :

   a) What do you mean by target identification step in process of drug discovery and development?

   b) What is analog?

   c) What do you mean by serendipitous way of drug discovery?

   d) Define Taft's steric constant.

   e) What is dummy variable?

   f) What do you mean by drug likeliness?

   g) What is flexible docking?

   h) Comment on role of bioinformatics in drug designing.

   i) Name any two databases used in ADMET profiling on NCE.

   j) Define local energy minima.

SECTION-B

2. Give detail account of bioisosteric based analog designing citing at least one case study.

3. What are the differences between CoMFA and CoMSIA? Give detail about contour analysis in both these 3D-QSAR approaches.

4. What is the difference between docking and pharmacophore-based screening? Give the advantages and disadvantages each of these two CADD techniques.

SECTION-C

5. Describe the clinical observation-based lead optimization with suitable examples.

6. Describe the additivity model of conventional QSAR along with advantages and disadvantages.

7. Describe Lipinski's rule of five and its importance in *in silico* drug design.

8. Give outline for pharmacophore mapping methodology.

9. Describe the ADMET databases which are commonly used for in *silico* drug designing.

10. Describe the concept of quantum mechanics and its utility in molecular modeling.

11. Discuss the principle, advantages and disadvantages of conjugate gradient.

12. Explain the role of chemo-informatics in drug designing.

13. Discuss relationship between Hansh and Free Wilson models.

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NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.